Exponential type

When we talked earlier about the definition of the Laplace transform, we worked an example to find the Laplace transform of $f(t) = e^{-2t}$, and we found

$$F(s) = \frac{1}{s+2} \qquad \text{given } s > -2$$

Without this restriction on the value of s, the Laplace integral diverges, and the transform $F(s)$ is therefore undefined.

At this point, we want to define more clearly some conditions that are sufficient for guaranteeing the existence of the Laplace transform.

Specifically, if we can show that a function $f(t)$ 1) is piecewise continuous on $[0,\infty)$ and 2) of exponential order, then we know that the Laplace transform $\mathscr{L}(f(t)) = F(s)$ exists for $s > \alpha$.

These two conditions aren't necessary for a Laplace transform to be defined. There are some functions that aren't piecewise continuous, but still have a defined Laplace transform. And there are some functions that aren't of exponential order, but still have a defined Laplace transform.

So these conditions aren't necessary, but they are sufficient. If a function meets these two conditions, then we know its Laplace transform exists. It's helpful to know that these two conditions are sufficient for the existence of the Laplace transform, because most of the functions we'll deal with throughout these transform lessons meet both of these conditions.

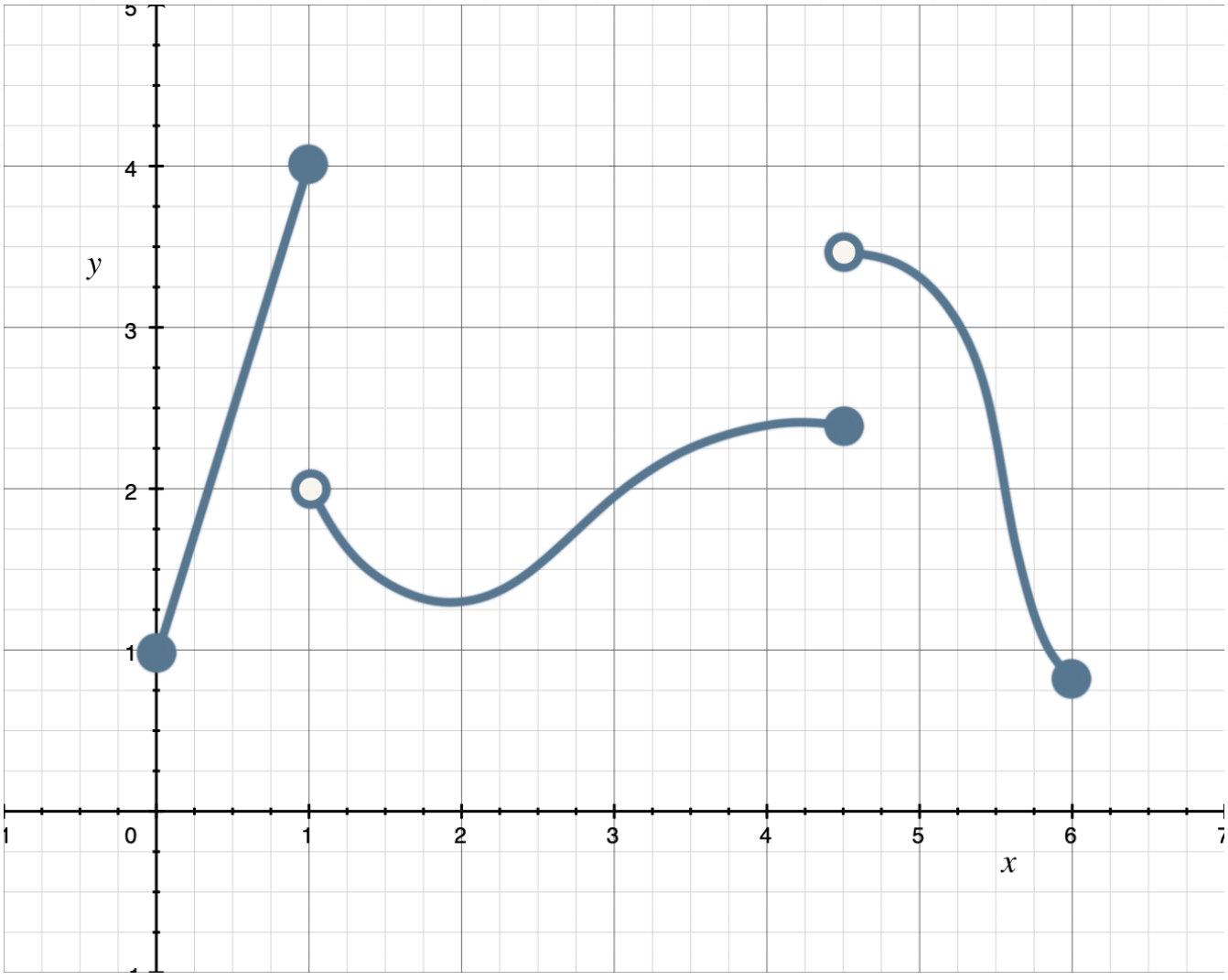
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Piecewise continuous functions

A function is **piecewise continuous** on $[0,\infty)$ if it includes a finite number of finite discontinuities (none of the discontinuities are infinite discontinuities, like at a vertical asymptote), and is otherwise continuous between these finite discontinuities.

Here's an example of a piecewise continuous function:



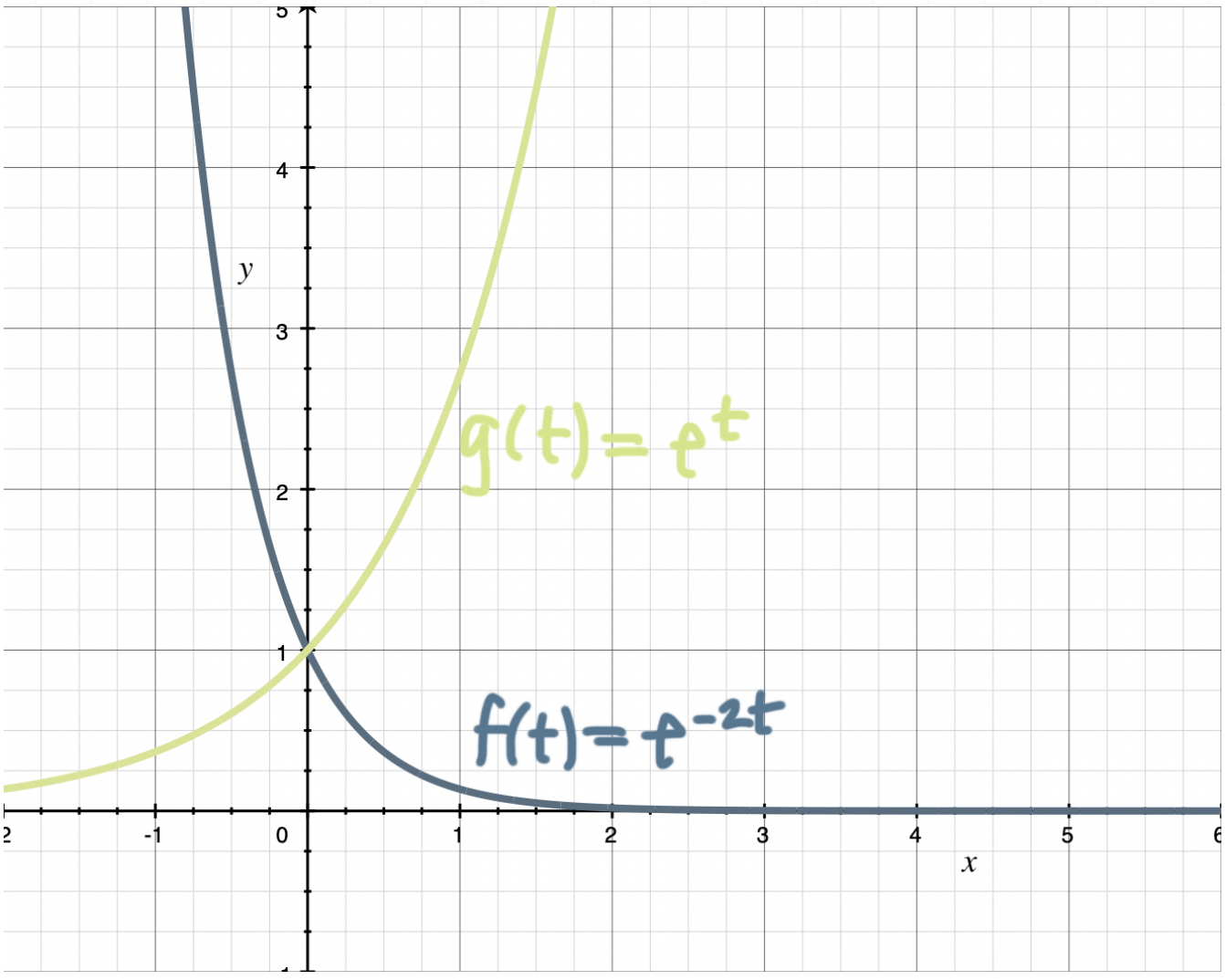
Exponential type (order)

Formally, we say that a function f is of **exponential type**, or of **exponential order** α, if there exist constants α, $M > 0$, $T > 0$ such that $|f(t)| \leq Me^{\alpha t}$ for all $t > T$.

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But really all this means is that $f(t)$ is of exponential type if we can find any $e^{\alpha t}$ that grows faster than $f(t)$, causing the value of the fraction $f(t)/e^{\alpha t}$ to converge as t gets infinitely large.

We started this lesson with the function $f(t) = e^{-2t}$. We know this function is of exponential type because the function $g(t) = e^{t}$ grows faster than $f(t) = e^{-2t}$.



In this case, we chose $\alpha = 1$, $M = 1$, and $T > 0$ for all $t > 0$. So

$$|f(t)| \leq Me^{\alpha t}$$

$$|e^{-2t}| \leq 1e^{1t}$$

$$e^{-2t} \leq e^{t}$$

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Because this inequality holds for all $t > 0$, we know $f(t) = e^{-2t}$ is of exponential type. Because we chose $\alpha = 1$, we can say that $f(t) = e^{-2t}$ is of exponential order 1, and we can conclude that the Laplace transform will exist for all $s > \alpha = 1$.

Let's do an example where classify functions as functions of exponential type or not of exponential type.

Example

Say whether or not each function is of exponential type.

$$f(t) = 2\cos t$$

$$g(t) = e^{-t}$$

$$h(t) = e^{t^2}$$

To determine whether a function is of exponential type, we'll try to find an exponential function that grows faster.

If we consider the function $f(t) = 2\cos t$, the value of $\cos t$ is never greater than 1, but $e^{\alpha t}$ grows exponentially for any $\alpha > 0$. So $e^{\alpha t}$ will grow infinitely large while $f(t) = 2\cos t$ remains small, so $f(t) = 2\cos t$ is of exponential type.

If we consider the function $g(t) = e^{-t}$, we know the function is actually decreasing, but $e^{\alpha t}$ grows exponentially for any $\alpha > 0$. So $e^{\alpha t}$ will grow infinitely large while $g(t) = e^{-t}$ gets smaller and smaller, so $g(t) = e^{-t}$ is of exponential type.

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If we consider the function $h(t) = e^{t^2}$, we can't find any $e^{\alpha t}$ that will grow faster than $h(t) = e^{t^2}$, because t^2 will always grow faster than αt, regardless of which positive constant we choose for α. Because αt is linear and t^2 is quadratic, eventually t^2 will always beat αt in a race to infinity. So $h(t) = e^{t^2}$ is not of exponential type.

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